RF BASED SMART AGRICULTURE MONITORING SYSTEM

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Arduino Tx/Rx + 433MHz RF Module | Presenter Name | Class | Date

SMART AGRI PROJECT

PROBLEM STATEMENT

Challenges in Traditional Agriculture

Manual monitoring is time-consuming and inefficient for farmers. Internet and Wi-Fi connectivity remains unavailable in many rural farming areas across India. Delayed alerts about soil conditions, temperature changes, or gas leaks lead to significant crop damage and financial losses for farming communities.

PROJECT MAIN IDEA

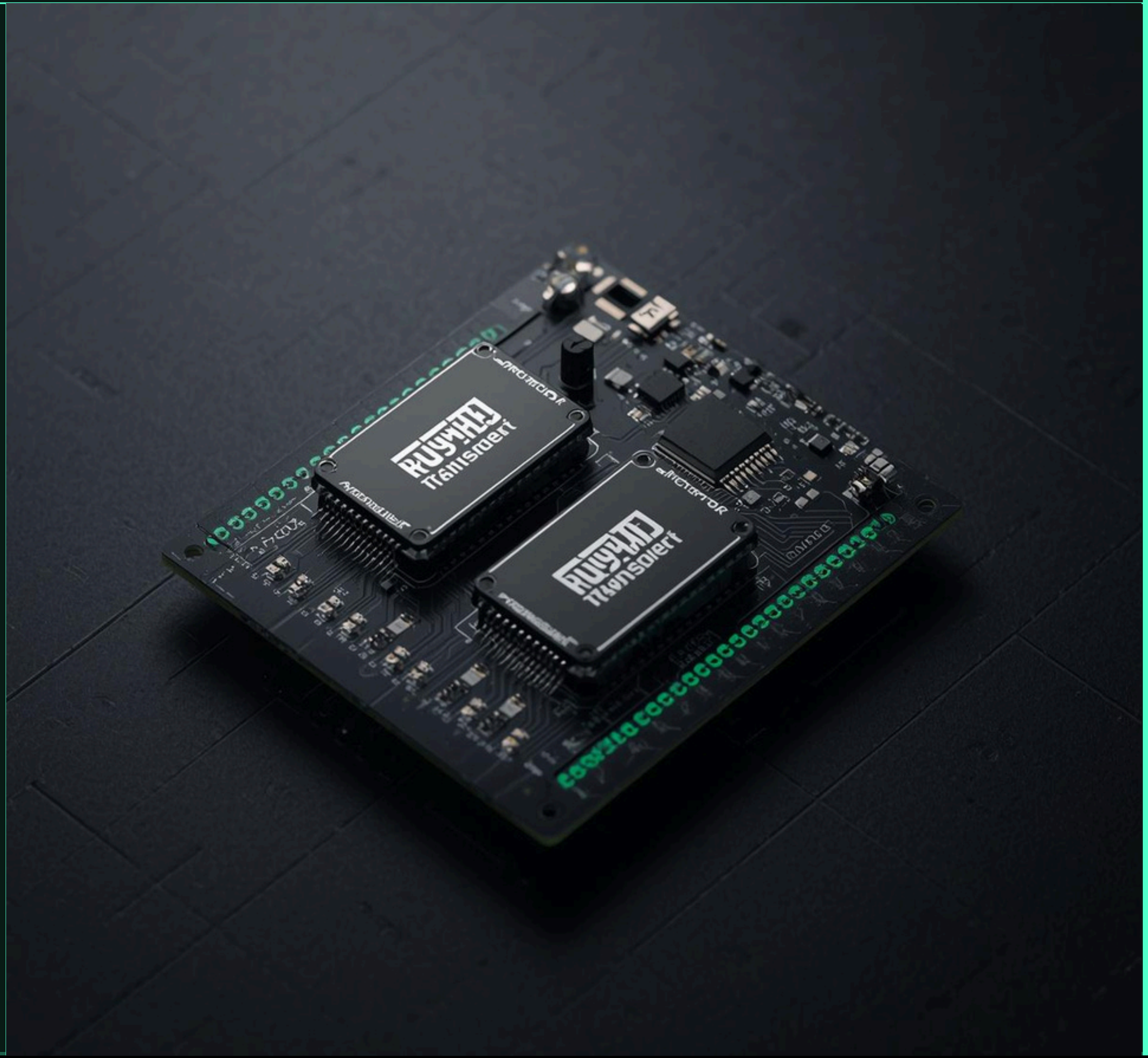
Wireless Offline Agriculture Monitoring

Our system operates on a simple two-node architecture: a Field Unit (Transmitter) deployed in the farm collects real-time sensor data, while a Home Unit (Receiver) displays readings and triggers alerts. Data is transmitted wirelessly via 433MHz RF modules—no internet or Wi-Fi required. This enables farmers to remotely monitor soil moisture, temperature, humidity, and gas levels in real-time, receiving early warnings to protect their crops.

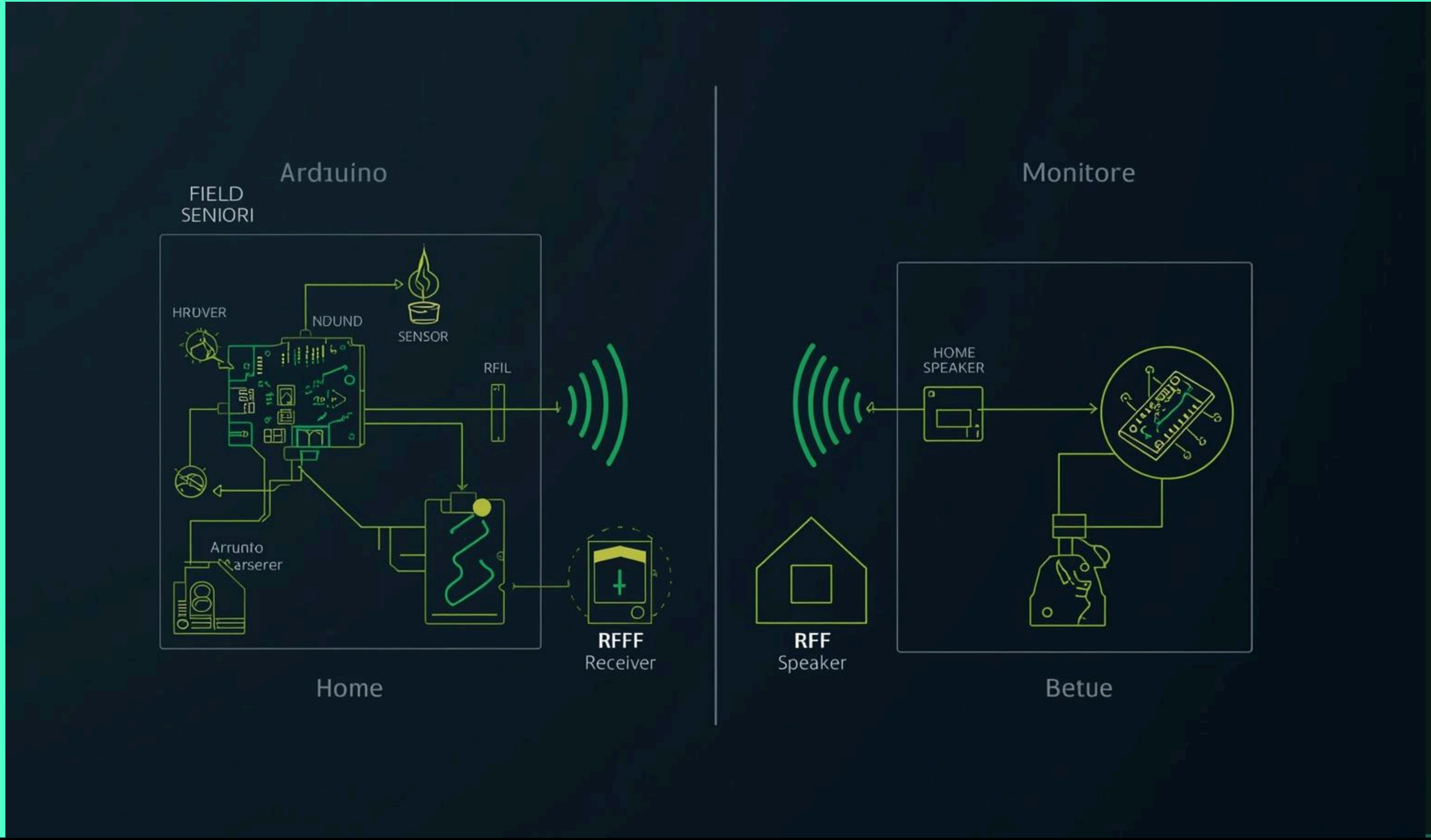
WHY USE 433MHZ RF COMMUNICATION?

The Smart Choice for Rural Farming

• Low cost and widely available modules — affordable for small-scale farmers
• Longer transmission range ideal for expansive rural farm fields
• Simple setup without complex network infrastructure or technical expertise
• Reliable offline communication — no internet or Wi-Fi dependency
• Robust signal penetration through obstacles and vegetation



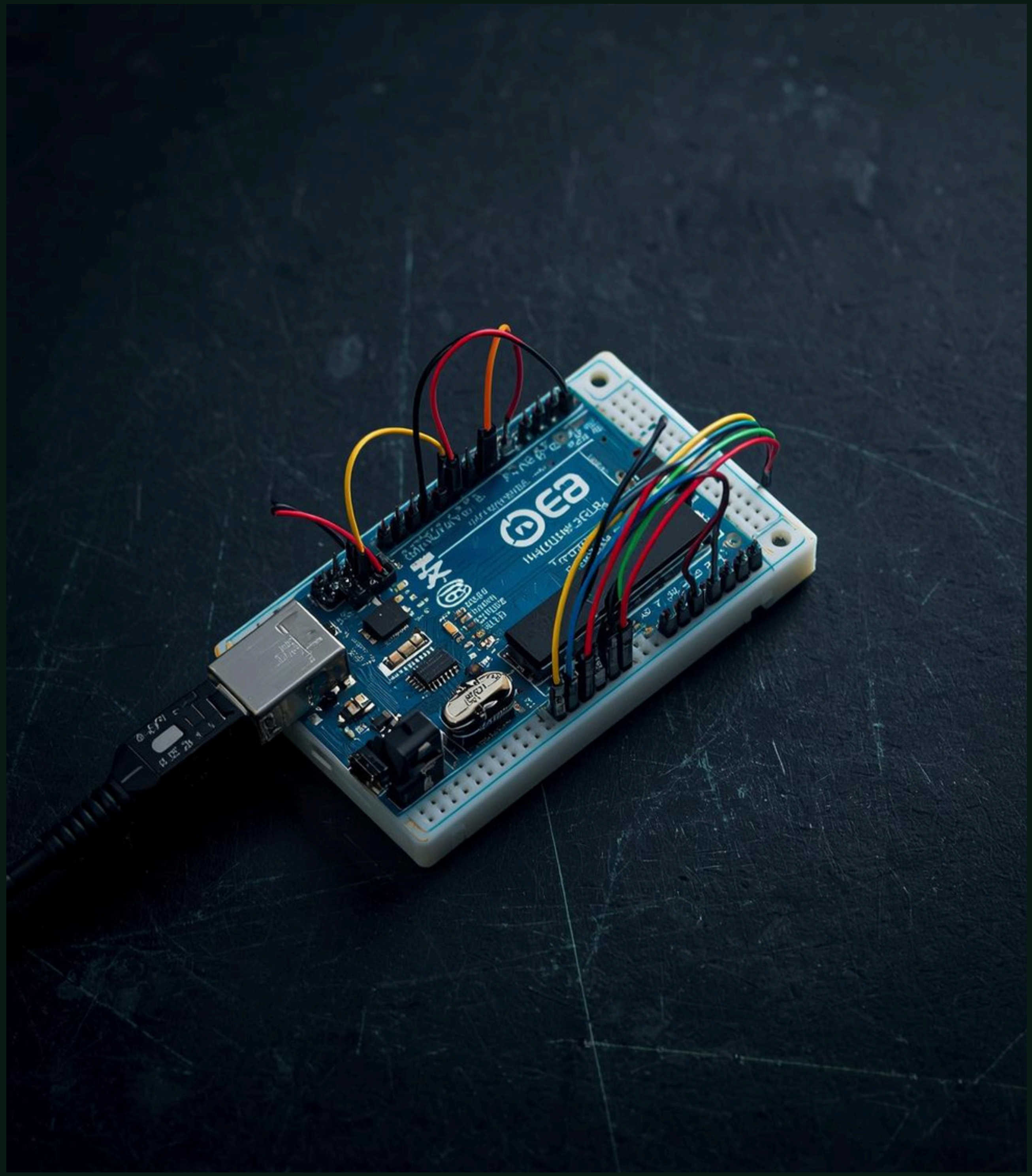
SYSTEM ARCHITECTURE OVERVIEW



Two-Node Wireless Communication

Field Unit: Arduino Uno + Sensors (Soil Moisture, DHT-22, MQ-5, MQ-135) + 433MHz RF Transmitter sends data wirelessly to Home Unit: Arduino Uno + RF Receiver + Buzzer + Serial Monitor for real-time alerts.

FIELD UNIT (TRANSMITTER)



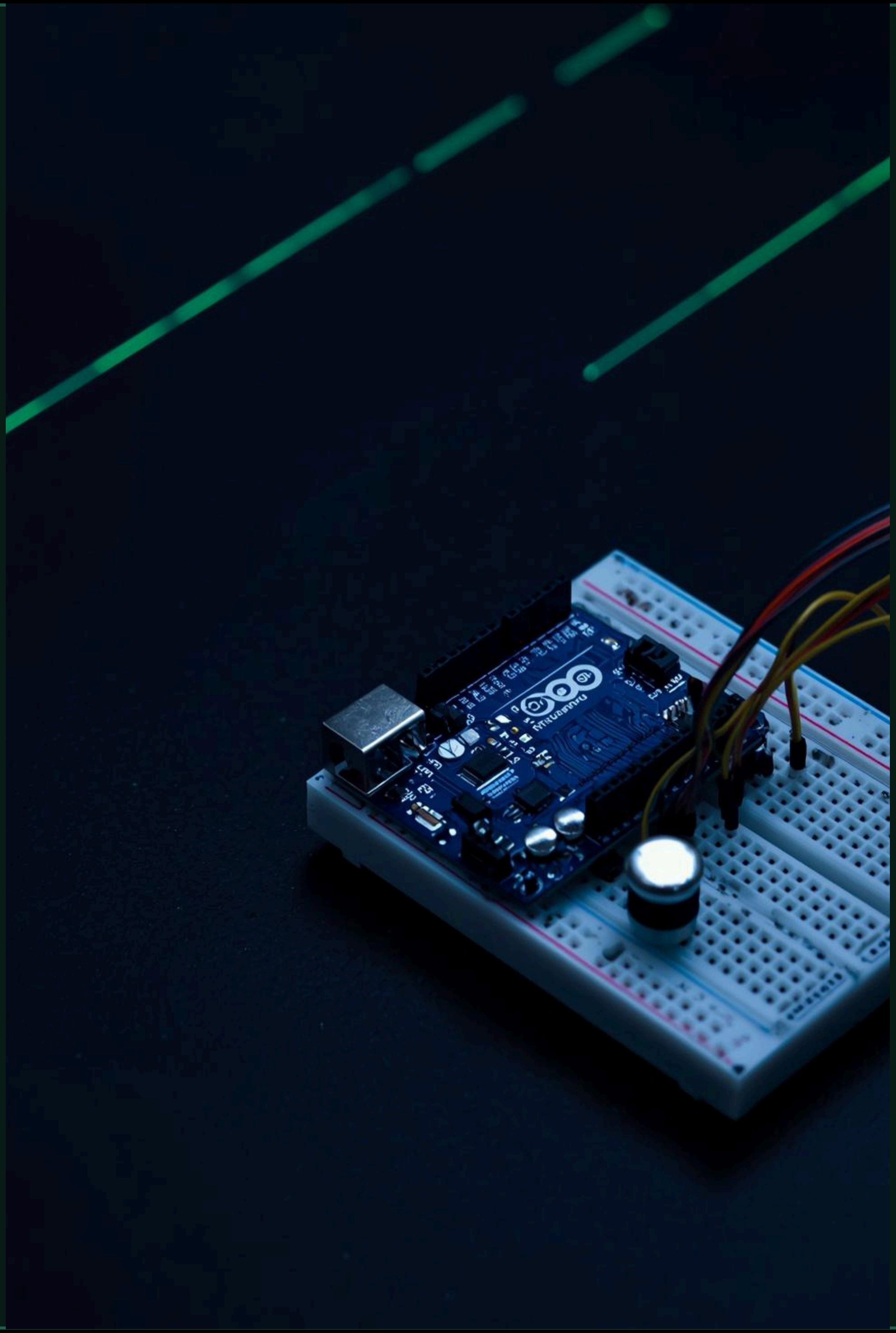
Components & Functions at Field Side

Arduino Uno collects real-time sensor data from the field. Sensors include Soil Moisture (YL-69), Gas Detection (MQ-5), Air Quality (MQ-135), and Temperature & Humidity (DHT-22). Data is formatted and transmitted wirelessly via 433MHz RF module.

HOME UNIT (RECEIVER) DETAILS

Command Centre at Your Doorstep

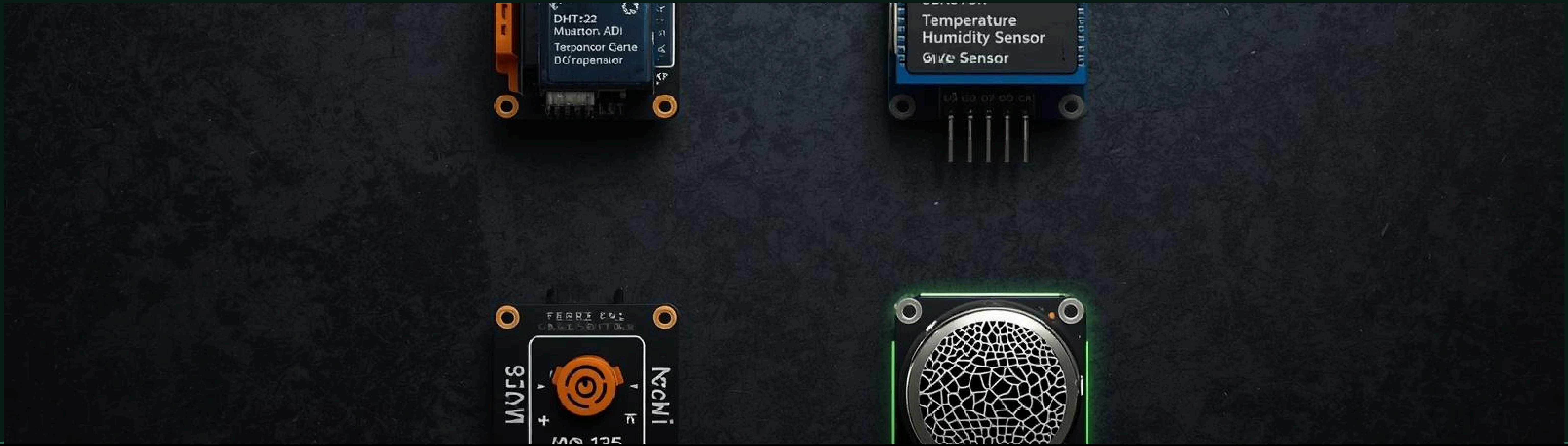
The Home Unit serves as the farmer's monitoring hub, built around an Arduino Uno microcontroller. It receives wireless data packets via the 433MHz RF receiver module, decodes sensor values in real-time, and displays readings on a Serial Monitor connected to a laptop or PC. When critical thresholds are breached—such as low soil moisture or high gas levels—a buzzer activates instantly, alerting the farmer to take immediate action without needing internet connectivity.



SENSORS USED IN THE SYSTEM

Smart Monitoring Components

YL-69 Soil Moisture Sensor: Detects soil dryness for irrigation decisions. DHT-22: Measures temperature & humidity for crop environment. MQ-5 Gas Sensor: Detects flammable gases and fire risks. MQ-135 Air Quality Sensor: Monitors harmful gases affecting crops.



SOIL MOISTURE SENSOR (YL-69)

Intelligent Irrigation Decision Support

The YL-69 soil moisture sensor is the foundation of smart irrigation management. It detects soil dryness or wetness by measuring electrical conductivity between two probes inserted into the soil.

• Dry soil = High resistance = Low output value
• Wet soil = Low resistance = High output value

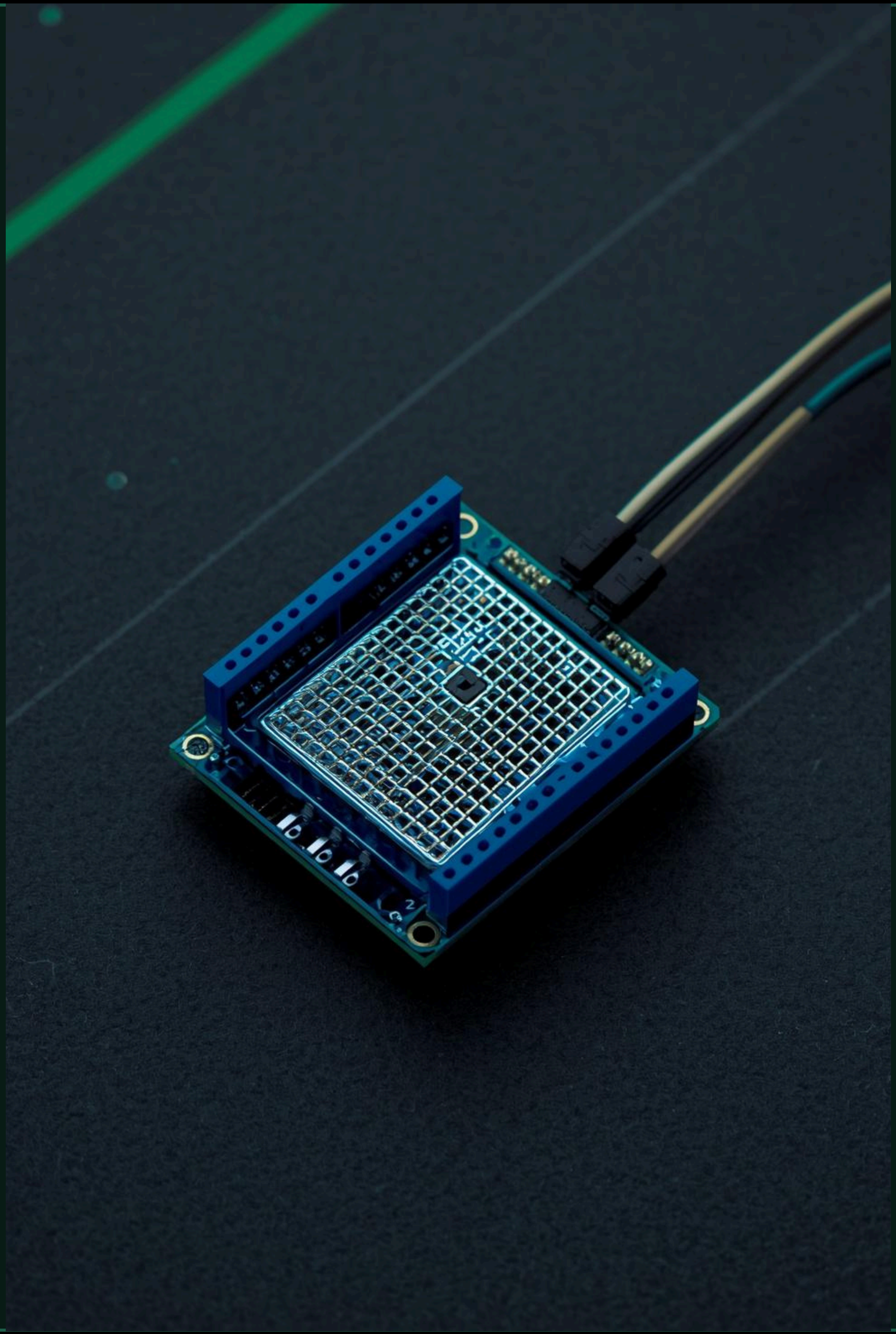
Threshold-based alerts automatically trigger when moisture drops below critical levels, enabling farmers to make timely irrigation decisions and prevent crop stress or water wastage.

DHT-22 SENSOR

Temperature & Humidity Monitoring

The DHT-22 sensor measures ambient temperature and humidity with high precision, providing critical environmental data for crop monitoring.

- Temperature Range: -40°C to 80°C (±0.5°C accuracy)
- Humidity Range: 0-100% RH (±2-5% accuracy)
- Helps monitor crop environment conditions in real-time
- Enables data-driven irrigation and ventilation decisions
- Low power consumption ideal for field deployment



MQ-5 GAS SENSOR

Safety Monitoring & Fire Detection

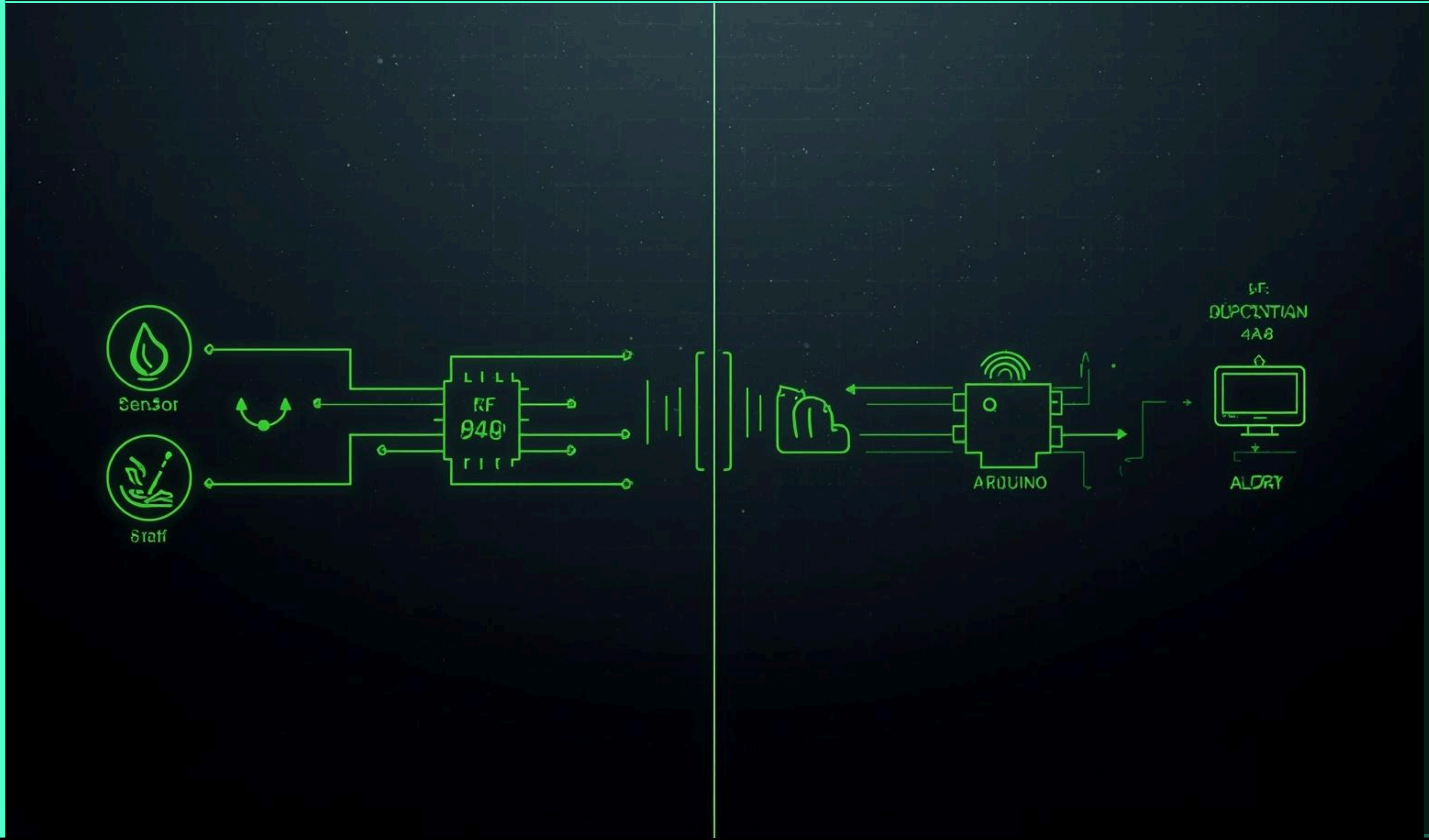
The MQ-5 sensor plays a critical role in farm safety by detecting flammable gases including LPG, natural gas, and combustible vapours. When gas levels exceed safe thresholds, the system triggers an immediate buzzer alert at the home unit, enabling rapid response to potential fire hazards. This proactive detection protects both crops and farmers from dangerous situations.

MQ-135 AIR QUALITY SENSOR

Monitoring Environment Quality

The MQ-135 sensor detects harmful gases including ammonia, benzene, alcohol, and smoke particles that affect crop health. It continuously monitors pollution levels in the field environment, triggering immediate alerts when unsafe conditions are detected—protecting both crops and farmer wellbeing.

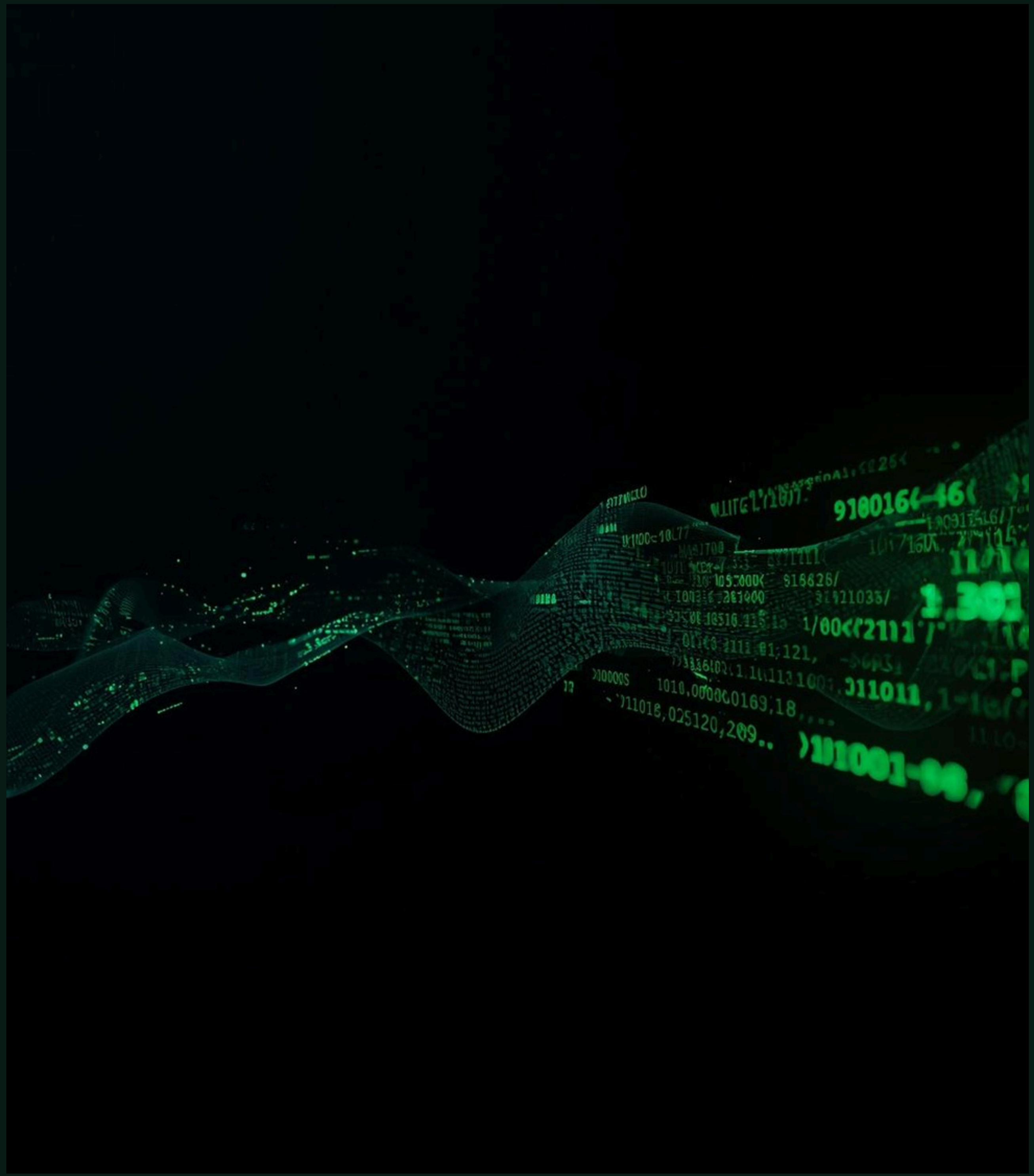
DATA FLOW IN THE SYSTEM



From Sensing to Smart Alerts

Sensors read environmental data → Arduino formats into data packets → RF Transmitter sends via 433MHz → RF Receiver captures packets → Arduino decodes values → Serial Monitor displays readings + Buzzer alerts on critical thresholds

DATA PACKET FORMAT EXAMPLE



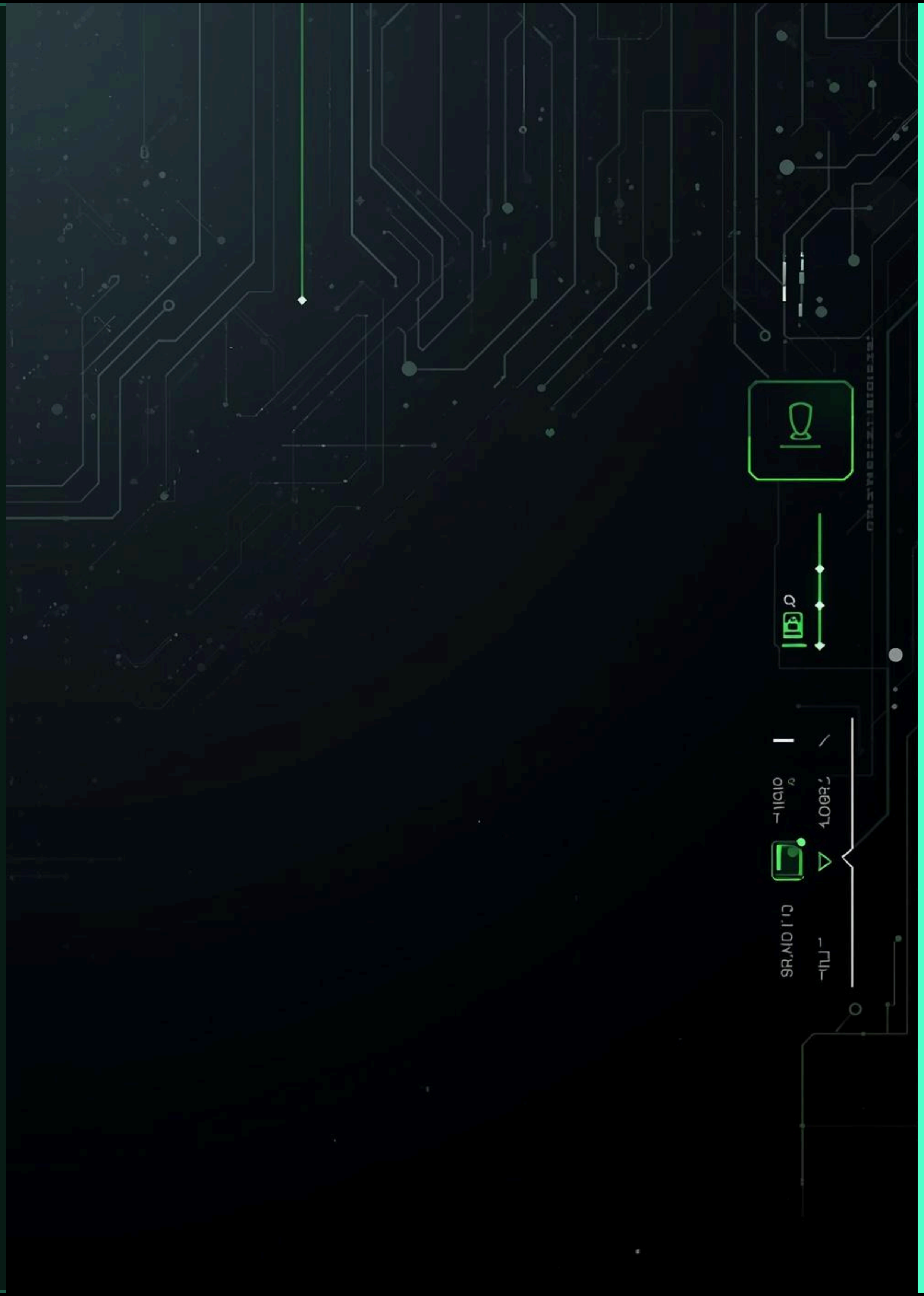
Communication Data Structure

Example string: M:450 T:29 H:60 G5:320 G135:410
M = Soil Moisture | T = Temperature (°C) | H = Humidity (%)
G5 = MQ-5 Gas Value | G135 = MQ-135 Air Quality Value
Simple, comma-free format ensures reliable RF transmission and easy parsing at the receiver end.

ALERT LOGIC

Smart Trigger Conditions

The system activates buzzer alerts based on critical sensor thresholds:
• Soil Moisture Below Threshold → Alert ON
Indicates dry soil requiring immediate irrigation
• MQ-5 Gas Reading Above Safe Level → Alert ON
Detects flammable gases or potential fire risk
• MQ-135 Air Quality Above Safe Level → Alert ON
Warns of harmful gases affecting crop health
• All Parameters Normal → Alert OFF
System continues silent monitoring



OUTPUT AND DEMO VISUALIZATION

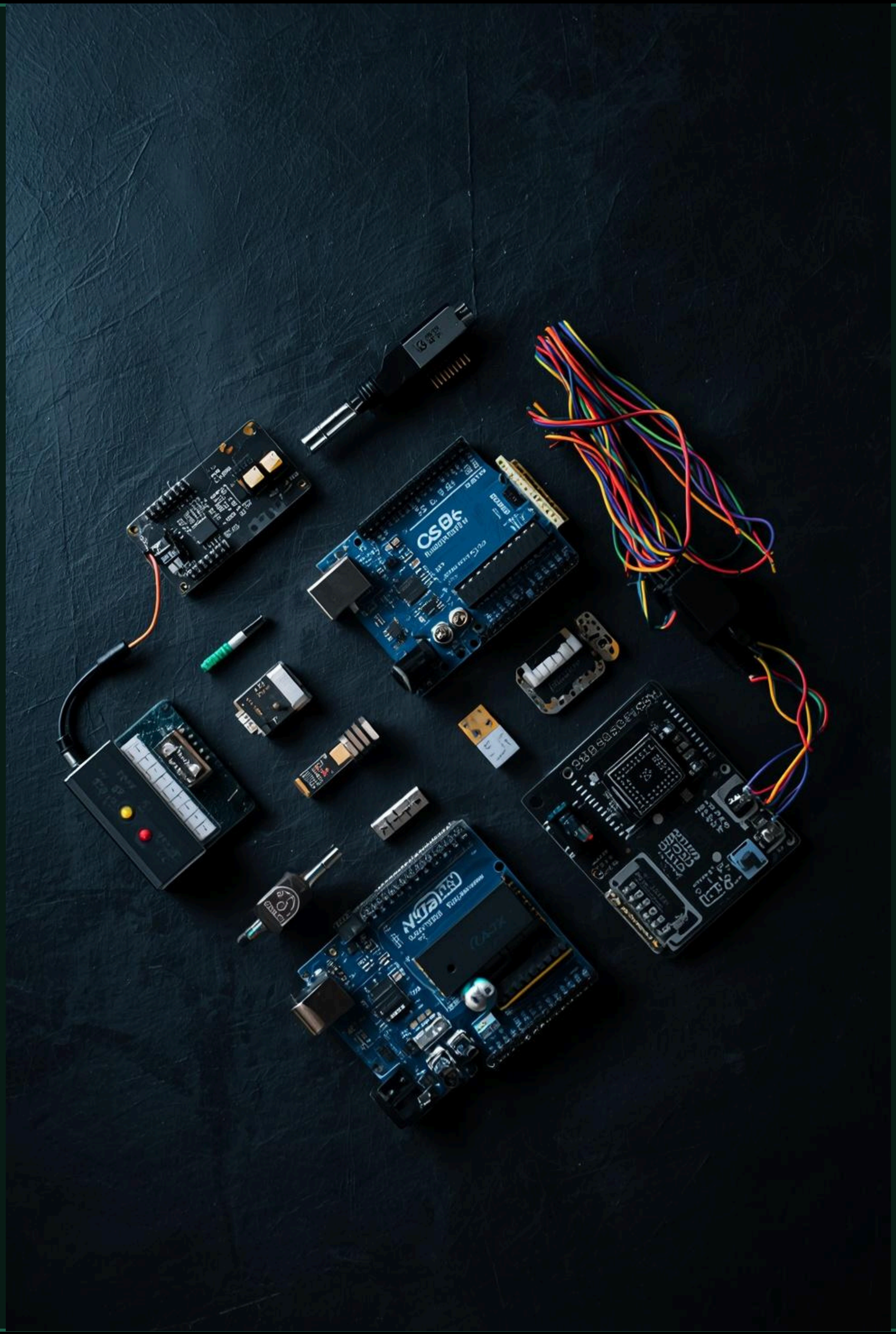
What the Farmer Sees at Home

Live sensor readings displayed on Serial Monitor in real-time. Buzzer sounds immediately on emergency conditions like low soil moisture or high gas levels. Real-time monitoring dramatically improves response time, enabling farmers to take quick action and protect their crops from damage.

COMPONENTS LIST (SHOPPING LIST)

Required Hardware

- 2× Arduino Uno boards
- 1 pair 433MHz RF Tx/Rx modules
- MQ-5 Gas Sensor
- MQ-135 Air Quality Sensor
- DHT-22 Temperature/Humidity Sensor
- YL-69 Soil Moisture Sensor
- Buzzer, breadboards, jumper wires



ADVANTAGES OF THE SYSTEM

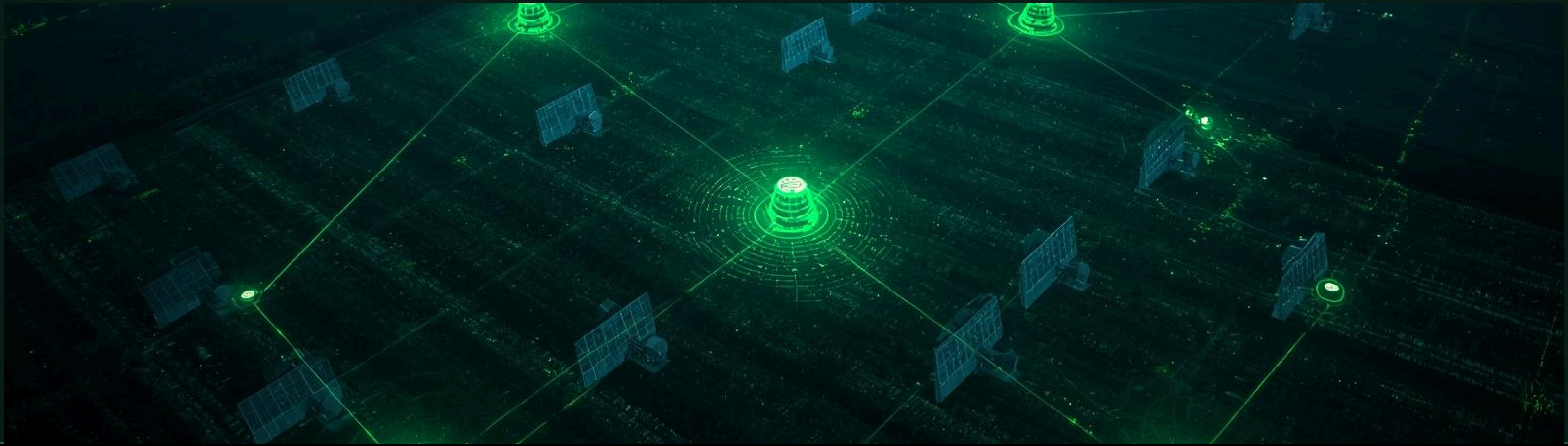
Key Benefits Highlighted

- Works without internet or Wi-Fi connectivity
- Low cost and easy to implement in rural areas
- Enables remote, real-time monitoring of field conditions
- Quick alerts reduce crop loss and increase farmer safety
- Simple setup requires no complex network infrastructure

APPLICATIONS AND FUTURE SCOPE

Expanding the System's Potential

Multi-node networks for larger farm coverage • Add LCD/OLED display for local viewing • Integrate GSM module for SMS alerts • Use solar power for remote unit energy • Upgrade to IoT platforms (ESP32, etc.)



THANK YOU

Questions & Discussion Welcome

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